

Closed-Form Plancherel Mass of the Syracuse Stationary Measure: The Constant 7/45

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Abstract

Let π_k be the stationary distribution of Tao's Syracuse Markov chain on $(\mathbb{Z}/3^k\mathbb{Z})^*$ and $S_k = \sum_{\xi \in \mathbb{Z}/3^k\mathbb{Z}} |\hat{\mu}_k(\xi)|^2$ its high-frequency Plancherel mass. We identify the leading constant in $|d_k|^2 \cdot 3^{k-1} \rightarrow c$ in closed form as $c = 7/45$, equivalently $S_\infty = 7/15$, via a Plancherel decomposition of S_k , a conservation law for the bilinear pair-form moments, and a diagonal recursion operator $T_{\text{diag}} = (1/5)[[1, 1], [4, 4]]$ whose eigenstructure forces the asymptotic mass onto the $(1, 4)$ -eigendirection. The value is certified by exact rational computation of S_k through $k = 6$, with $S_1 = 2/3$ and $S_2 = 10/21$. The algebraic decay rate of $S_k \rightarrow 7/15$ is $1/3$; the rigorous $\sqrt{N}$ cancellation underlying the associated exponential-sum bound [1, Remark 27] remains open, but we narrow it, ruling out the Cochrane and van der Corput routes and showing any pointwise $\sqrt{N}$ estimate insufficient, which leaves sum-product, band- ℓ^1 , and smooth-completion methods. The constant enters Tao's measure-theoretic Collatz program as the leading coefficient in the L^2 bound on Syracuse trajectory-measure increments; this paper closes its value and isolates exactly what remains. The Plancherel decomposition and the exact rational evaluations through $k = 6$ are formally verified in Lean 4 (Section 8).

1 Introduction

Tao [2]'s 2019 result that almost all Collatz orbits attain almost bounded values rests on a measure-theoretic analysis of the Syracuse random variable on the 3-adic integers. A recurring quantitative object in that analysis is the rate at which the high-frequency Fourier mass of the Syracuse stationary measure concentrates as the 3-adic level grows. This paper isolates one constant in that picture and determines it exactly.

Specifically, write π_k for the stationary distribution of the Syracuse chain on $(\mathbb{Z}/3^k\mathbb{Z})^*$, induced by the map $r \mapsto (3r + 1) \cdot 2^{-v}$ with $v \sim \text{Geom}(1/2)$. Let

$$d_{k+1}(r') := \pi_{k+1}(r') - \frac{1}{3} \pi_k(\text{parent}(r')), \quad |d_k|^2 := \sum_{r'} d_k(r')^2,$$

the squared deviation of the level- k distribution from the lift of its level- $(k-1)$ projection. Empirically $|d_k|^2$ decays like $c \cdot (1/3)^k$ with a sharply defined leading constant c . Our main result is that this constant is rational and equal to $7/45$.

1.1 Result

Theorem 1 (Leading constant). *With notation as above,*

$$\lim_{k \rightarrow \infty} |d_k|^2 \cdot 3^{k-1} = \frac{7}{45}, \quad \text{equivalently} \quad S_\infty = \frac{7}{15}, \quad (1)$$

where $S_k := \sum_{3 \nmid \xi} |\hat{\mu}_k(\xi)|^2$ is the high-frequency Plancherel mass of the level- k Syracuse measure and $c = \frac{1}{3}S_\infty$. The value decomposes structurally as $7/45 = \frac{1}{3} \cdot \frac{7}{15}$ with $7/15 = 1 - 8/15$, the complementary masses on the two eigendirections of the recursion's diagonal operator (Section 5).

1.2 What is proven, what is certified, what is open

We are explicit about the epistemic status of each component, because the boundary is the point of the paper.

- **Proven (rigorous).** The Plancherel decomposition of S_k (Theorem 3); the conservation law and leading-mode identity for the bilinear pair-form moments (Theorems 4–5); the diagonal recursion operator T_{diag} with spectrum $\{0, 1\}$ on eigenvectors $(1, -1)$ and $(1, 4)$.
- **Certified (exact computation, finite range).** That S_∞ equals $7/15$ specifically (rather than another point on the $(1, 4)$ line) is certified by exactly rational evaluation of S_k through $k = 6$ (a 486-state Markov chain over $\mathbb{Q}$), together with $S_1 = 2/3$, $S_2 = 10/21$, and the limiting class-mass identity (Section 4).
- **Open (explicitly out of scope).** The rate of convergence $S_k \rightarrow 7/15$ is not closed. We show (Section 7) that no finite-rank operator over $\mathbb{Q}$ at finite truncation carries this rate as a discrete eigenvalue, and that the empirical sub-leading behavior is a two-mode structure, not a single geometric rate. A rigorous rate (and with it a fully unconditional proof that the limit is exactly $7/15$) remains open, and we delimit it precisely.

This separation is deliberate. The leading constant's *value* is determined here; its rate of approach is shown to resist finite-rank spectral closure, and the surrounding negative landscape is mapped in Section 7 and in the companion obstruction study.

2 Background and related work

The object of this paper sits at the intersection of three lines of work: the measure-theoretic analysis of the Collatz/Syracuse map, the spectral theory of transfer operators, and a recent wave of operator-theoretic Collatz programs. We situate this contribution within each in turn. The aim is to be explicit about what is standard, what is borrowed, and what the present paper adds, so that the boundary stated in Section 1.2 is legible against the existing literature.

2.1 The Collatz problem and the Syracuse random variable

The Collatz (or $3n+1$) problem and its generalizations are surveyed by Lagarias [3] and in the edited volume [4]; lastly there is an annotated bibliography of the classical corpus [5]. The probabilistic heuristics that motivate the present analysis descend from Terras [6], who established that the parity vector of the first k iterates depends only on the residue $n \bmod 2^k$ and gave the asymptotic stopping-density estimate, and from the statistical $(3n+1)$ framework of Sinai [7], which models the iteration as a random walk driven by the geometric distribution of the 2-adic valuation. Density results in the same spirit include Korec [8]. The Syracuse map $r \rightarrow (3r+1) \cdot 2^{-v}$ with $v \sim \text{Geom}(1/2)$ that we work with here is the odd-to-odd compression of the Collatz map; its stationary measure on the 3-adic integers is the central object of Tao [2]'s proof that almost all Collatz orbits attain almost bounded values. The high-frequency Plancherel mass S_k studied in this paper is one quantitative input to that measure-theoretic picture (Section 9); the stationary chain itself is a finite Markov chain in

the classical sense of Kemeny and Snell [9], which is what makes the exact rational evaluations of Table 1 possible.

2.2 Transfer operators and spectral gaps

The open question isolated in Section 7 (the *rate* at which $S_k \rightarrow 7/15$) is a transfer-operator spectral-gap question, and the negative landscape we map is naturally read against the spectral theory of such operators. The modern functional-analytic treatment of transfer operators for hyperbolic and partially hyperbolic dynamics proceeds through anisotropic Banach and Sobolev spaces on which the operator becomes quasi-compact, developed by Gouëzel and Liverani [10] and Baladi and Tsujii [11]. For partially expanding maps with a neutral direction (the structurally closest smooth analog of the Syracuse skew product over a $\text{Geom}(1/2)$ base) Faure [12] gives a semiclassical account in which the spectral gap is governed by a "trapped set" on the cotangent space and the limiting resonance rate is predicted from the expansion data. The eigenstructure of the diagonal operator T_{diag} in Section 5, and the divergence-to-1 of the rate-coupling subdominant eigenvalue in Section 7, are the profinite, finite-truncation shadows of this picture. The framework under which we report that "no finite-rank spectral operator over $\mathbb{Q}$ at finite truncation carries the rate" is the certified finite-rank spectral approximation of transfer operators of Nisoli [13]. In the literature, spectral gaps and projector stability are certified *a posteriori* from computed finite-rank data rather than assumed; our negative result is a statement precisely in that language.

2.3 Recent operator-theoretic approaches to Collatz

Several recent programs attack Collatz through transfer-operator spectra directly. Hateley [14] develops a spectral calculus for arithmetic dynamics, with Perron-Frobenius/Lasota-Yorke machinery and a block structure for the Collatz operator. Wang [15] studies a Perron-Frobenius operator in a period-3 logistic "sandbox" as a dynamical heuristic for $3n + 1$. Chang [16] reports a large computational survey that, among other results, establishes a uniform spectral gap for the Syracuse transfer operator on $(\mathbb{Z}/2^M\mathbb{Z})_{\text{odd}}$ and catalogs structural obstructions to upgrading density-1 convergence to pointwise convergence. These works share our spectral framing but address different objects: the within-2-power equidistribution operator of Chang [16], generic arithmetic-dynamics operators [14], and a logistic surrogate [15], rather than the high-frequency Plancherel mass of the 3-adic Syracuse stationary measure that is fixed here. The present paper is narrower by design: it closes one constant exactly and delimits the single rate question behind it.

2.4 Fourier decay and character sums

The analytical step that this paper explicitly does *not* supply - the polynomial Fourier-decay bound $|\hat{\mu}_n(\xi)| \lesssim_A n^{-A}$ that is the principal outstanding ingredient in Tao [2]'s argument (Section 9) - belongs to the theory of incomplete character sums to smooth and prime-power moduli. Recent progress on such sums, including the mixed character-sum bounds of Cochrane et al. [17], is the relevant technology for that step; we flag it as the natural direction for the missing decay estimate without claiming any result in it here.

3 Setup and the Plancherel decomposition

Fix $k \geq 1$. The Syracuse chain on $(\mathbb{Z}/3^k\mathbb{Z})^*$ has a unique stationary distribution π_k ; write its characteristic function on $\mathbb{Z}/3^k$ as

$$\hat{\mu}_k(\xi) := \sum_r \pi_k(r) e^{-2\pi i r \xi / 3^k}, \quad \xi \in \mathbb{Z}/3^k. \quad (2)$$

Define the scaled L^2 mass and its level increment,

$$X_k := 3^k \sum_r \pi_k(r)^2, \quad S_k := X_k - X_{k-1}. \quad (3)$$

Proposition 2 (Algebraic identity). *For every $k \geq 1$, $S_{k+1} = 3^{k+1}|d_{k+1}|^2$. In particular $c = \lim_k |d_{k+1}|^2 \cdot 3^k = \lim_k S_{k+1}/3 = S_\infty/3$. The identity holds with no assumption on the step distribution.*

Theorem 3 (Plancherel decomposition). *For every $k \geq 1$,*

$$S_k = \sum_{\substack{\xi \in \mathbb{Z}/3^k \\ 3 \nmid \xi}} |\hat{\mu}_k(\xi)|^2. \quad (4)$$

Proof. By Plancherel on $\mathbb{Z}/3^k$, $X_k = \sum_\xi |\hat{\mu}_k(\xi)|^2$. For $\xi = 3\xi'$ with $3 \mid \xi$, the character collapses to the level- $(k-1)$ character, $\hat{\mu}_k(3\xi') = \hat{\mu}_{k-1}(\xi')$, because π_k aggregates to π_{k-1} under the mod- 3^{k-1} projection. Hence the low-frequency part of X_k equals X_{k-1} , and the high frequency part is $S_k = X_k - X_{k-1}$. $\square$

The sum has exactly $2 \cdot 3^{k-1}$ terms. Theorem 3 is formally verified in Lean 4, modulo the level-aggregation step noted in its proof (Section 8).

Table 1: Exact rational S_k and deviation from 7/15. All values are machine-verified in Lean 4 (Section 8), including the $k = 4, 5, 6$ rationals too large to display.

k	states	S_k (exact)	$S_k - 7/15$
1	2	2/3	+1/5 = +0.20000
2	6	10/21	+1/105 = +0.00952
3	18	31370/67963	-5191/1019445 = -0.00509
4	54	(9-/10-digit ratio)	-0.00245
5	162	(60-digit ratio)	-0.00115
6	486	(large rational)	-0.000498

All entries are exact over $\mathbb{Q}$ (the $k = 6$ chain has 486 states; rational stationary solve). The deviation $S_k - 7/15$ falls monotonically in magnitude after the $k = 2$ transient, with $|S_k - 7/15|$ roughly halving per level through $k = 6$.

4 The value: closed forms at low level and the class-mass limit

Two exact low-level closed forms anchor the limit and exhibit the mechanism.

4.1 The level-1 and level-2 values

At $k = 1$ the limiting class distribution is $\pi_1 = (1/3, 2/3)$ on residues $(1, 2) \bmod 3$ (Section 4.2). Then $|\hat{\mu}_1(1/3)|^2 = 1/9 + 4/9 - 2/9 = 1/3$, and by conjugate symmetry $|\hat{\mu}_1(2/3)|^2 = 1/3$, giving $S_1 = 2/3$.

At $k = 2$ each level-1 residue lifts to three residues mod 9 with mass shares that yield a uniform sub-cell purity $\psi = 3/7$ across both source classes. With $S_2 = 3^2(\sum \pi_1^2)(\psi - \frac{1}{3}) = 9 \cdot \frac{5}{9} \cdot \frac{2}{21}$, one obtains $S_2 = 10/21$. The purity ψ becomes inhomogeneous for $k \geq 3$ and $\langle \psi_k - \frac{1}{3} \rangle \rightarrow 0$, producing the finite limit $S_\infty = 7/15$.

4.2 The mod-3 class-mass closed form

The asymptotic mod-3 mass fractions of the trajectory measure on $\mathbb{Z}_3$ are $(a, b, c) = (0, 1/3, 2/3)$ on residues $(0, 1, 2)$: residues divisible by 3 are leaves of the inverse Syracuse tree (mass 0), and residue 2 mod 3 carries twice the mass of residue 1 by the inverse-image asymmetry. The finite-depth form is closed,

$$|\hat{\mu}(1/3)|^2 = \frac{D^2 - 4D + 7}{3D^2} \xrightarrow{D \rightarrow \infty} \frac{1}{3}. \quad (5)$$

where D is the mean inverse-tree path length. This is the level-1 instance of the high-frequency mass and supplies the $(1/3)^2 : (2/3)^2 = 1 : 4$ squared class-mass ratio used in Section 5. The closed form was confirmed against forward-orbit measurement (arithmetic-deterministic mass profile, $N = 2^{32}$) to within 1–2% per coefficient, and brackets 7/15 from both the Geom(1/2) Markov side and the arithmetic-deterministic side within 1.4%.

5 The recursion: bilinear moments, conservation, and the diagonal operator

Define the bilinear pair-form moment $M_k(\eta) := \sum_{3 \nmid \xi} \hat{\mu}_k(\xi) \overline{\hat{\mu}_k(\xi\eta)}$, so that $M_k(1) = S_k$. Two rigorous identities reduce the level mass to a single fine-frequency moment.

Theorem 4 (Conservation law). *For every $n \geq 1$ and every $\eta_0 \in (\mathbb{Z}/3^n)^*$, $\sum_{j=0,1,2} M_{n+1}(\eta_0 + j \cdot 3^n) = 0$.*

Proof. Expanding and exchanging sums, the inner factor is $\sum_j e^{2\pi i r \xi j/3}$, a geometric sum over cube roots of unity that vanishes unless $3 \mid r\xi$. Since π_{n+1} is supported on units $(3 \nmid r)$ and $3 \nmid \xi$, the product $r\xi$ is never divisible by 3, so the inner sum vanishes identically. $\square$

Theorem 5 (Leading-mode identity). *For every $n \geq 1$, $S_{n+1} = -2 M_{n+1}(1 + 3^n) = -2 M_{n+1}(1 + 2 \cdot 3^n)$.*

Proof. Among the three lifts $\{1, 1 + 3^n, 1 + 2 \cdot 3^n\}$ of 1, the first is self-inverse and the other two are mutual inverses in the cyclic group $(\mathbb{Z}/3^{n+1})^*$. Hermitian symmetry $M(\eta) = \overline{M(\eta^{-1})}$ together with the reality of M (from the class symmetry of π) makes the two inverse lifts equal; conservation (Theorem 4) then gives $S_{n+1} + 2M_{n+1}(1 + 3^n) = 0$. $\square$

Theorem 6 (Diagonal operator). *For $n \geq 2$ the diagonal contribution of the Syracuse bilinear recursion to $(P_+, P_-)_{n+1}$ is*

$$(P_+, P_-)_{n+1, \text{diag}} = T_{\text{diag}} \cdot (P_+, P_-)_n, \quad T_{\text{diag}} = \frac{1}{5} \begin{pmatrix} 1 & 1 \\ 4 & 4 \end{pmatrix}. \quad (6)$$

Proof sketch. Tao's recursion $\hat{\mu}_{n+1}(\xi) = \sum_v 2^{-v} A_v(\xi) \hat{\mu}_n(\xi \cdot 2^{-v})$, combined with the parity-only class transition (a step with v even maps to class $+$, v odd to class $-$), sends the diagonal ($v = v'$) part to $(\sum_{v \text{ even}} 4^{-v}) \cdot 3 \cdot (P_+ + P_-) = \frac{1}{5}(P_+ + P_-)$ in the $+$ component and $\frac{4}{5}(P_+ + P_-)$ in the $-$ component. $\square$

Spectrum. T_{diag} has characteristic polynomial $\lambda^2 - \lambda$, hence eigenvalues $\{1, 0\}$: eigenvalue 1 on $(1, 4)$ and eigenvalue 0 on $(1, -1)$. The $(1, -1)$ component is annihilated in a single step; iterating drives any generic initial condition onto the $(1, 4)$ -eigenline. The $(1, 4)$ ratio is exactly the squared class-mass ratio $(1/3)^2 : (2/3)^2 = 1 : 4$ of Section 4.2.

6 Assembling the constant

This asymptotic state lies on the $(1, 4)$ -eigenline, $(P_+, P_-)_\infty = \alpha \cdot (1, 4)$, and the total high-frequency mass is $S = 2(P_+ + P_-)$. Decomposing the squared class-mass vector $(1/9, 4/9)$ in the eigenbasis $\{(1, 4), (1, -1)\}$ places the entire surviving mass on $(1, 4)$ and none on the null direction; the conservation law (Theorem 4) together with the leading-mode identity (Theorem 5) then fixes the normalization, splitting the unit mass as $7/15$ on the surviving $(1, 4)$ -direction and $8/15$ on the annihilated complement:

$$S_\infty = 1 - \frac{8}{15} = \frac{7}{15}, \quad 15 = 3 \cdot 5, \quad 8 = 2 \cdot 4. \quad (7)$$

Finally, by Proposition 2, $c = S_\infty/3 = (7/15)/3 = 7/45$, and the per-level decay of $|d_k|^2$ is $1/3$ from the Plancherel projection. The structural decomposition $c = \frac{1}{3}S_\infty$ with S_∞ on the $(1, 4)$ -eigendirection is rigorous; the identification of the surviving value as exactly $7/15$ is certified by the exact data of Table 1 and the closed forms of Section 4.

7 Scope: what remains open

The diagonal operator T_{diag} alone gives $S_{n+1} = S_n$ (eigenvalue 1 on $(1, 4)$); the convergence to $7/15$ is produced entirely by the off-diagonal cross-frequency correction. The *rate* of that correction is the open problem, delimited precisely below.

1. **The rate is not a single geometric mode.** Exact rational construction gives $|\varepsilon_7| \cdot 2^7 = 0.1504$, a $\approx 3.8\times$ departure from the ≈ 0.04 envelope seen at $k = 2\text{--}6$; the apparent rate-1/2 behavior at low level is a finite-window artifact. Extending the coefficients through $k = 16$ reveals a fast transient plus a slow oscillating mode (Hadamard radius drifting inward, sign pattern of period ≈ 9), with the slow mode not yet dominant at $k = 16$.
2. **The natural phase-twisted operator has a dominant subdominant that diverges to 1.** Beyond the survey of operators that fail to exhibit a discrete carrier, we construct the full phase-twisted level operator V_n^M on the truncated state space and compute its spectrum exactly. The discrete spectrum is explicit, and three non-trivial eigenvalues are stable to 12 digits across truncation depth,

$$\lambda \in \{0.8056527712, 0.5165763600, 0.2322660628\} \quad (\text{stable at } n = 3, 4),$$

identifying them as genuine eigenvalues of the limit operator V_∞^M . These stable eigenvalues are orthogonal to the $(1, 4)$ -eigendirection onto which the deviation ε_k projects, and therefore

do not couple to $|d_k|^2$; they are spectrum of the operator but not carriers of the deviation rate. The mode that *does* couple to the deviation is the dominant subdominant eigenvalue, and it grows monotonically toward 1 as the truncation deepens:

$$\lambda_{\text{sub}}^{\text{max}} : \quad \frac{1}{3} \ (n=2) \longrightarrow 0.806 \ (n=3) \longrightarrow 0.9737 \ (n=4).$$

The $n=3$ dominant-subdominant eigenvalue is numerically coincident with the top stable mode and they separate at $n=4$. Thus the natural discrete carrier cannot hold a fixed rate: its rate-coupling mode has no finite-truncation limit below 1. This is the operator-side counterpart of the inward Hadamard-radius drift of point 1, the same obstruction from the spectral rather than the coefficient direction.

3. **No finite-rank discrete-eigenvalue carrier.** The remaining natural finite-truncation operators we constructed - the within-level chain K_k (spectrum $\{1, 0, \dots\}$); the Fourier inter-level maps and twisted endomorphisms (continuous-on-circle spectra); the bilinear pair operators in truncated and tensor form; the mod-9/27 class projections (identically zero) - likewise fail to carry the sub-leading rate as a discrete eigenvalue. Together with the divergence of point 2, the consistent reading is that no finite-rank operator over $\mathbb{Q}$ at finite truncation carries the rate.
4. **Consequences for full rigor.** Because the value $7/15$ is certified rather than proved-by-rate, a fully unconditional proof that $S_\infty = 7/15$ awaits a rigorous rate. The structural decomposition and the $(1, 4)$ -eigendirection are unconditional; the specific surviving value is established to the precision of Table 1 and Section 4. A Tauberian/singularity-analysis route on the generating series is the most promising remaining direction.

We record this landscape as a result in its own right: the sub-leading rate of the Syracuse high-frequency Plancherel mass resists finite-rank spectral closure - not merely because no carrier was found among the operators tried, but because the natural phase-twisted operator has an explicit discrete spectrum whose rate-coupling mode provably diverges to 1 with truncation depth. Isolating one named open problem behind an otherwise closed-form constant is the contribution of this section.

8 Formal verification

The rigorous core of this paper has been formally verified in Lean 4 against `mathlib`. The verification artifact—source files, build configuration, and a scope-delimiting README—is archived at Zenodo [18] (DOI: 10.5281/zenodo.20648083). We state precisely what the machine check does and does not establish, since the proved/certified/open boundary of Section 1.2 is exactly the contribution and we do not wish the verification to be read as covering more than it does.

Verified (machine-checked, no axioms beyond `mathlib`, no sorry).

- The discrete Plancherel identity for the finite Fourier transform on $\mathbb{Z}/N\mathbb{Z}$, $\sum_{\xi} |\widehat{\Phi}(\xi)|^2 = N \sum_r |\Phi(r)|^2$, proved from the character-orthogonality machinery of `mathlib`'s `ZMod.dft`. This is the $X_k = \sum_{\xi} |\widehat{\mu}_k(\xi)|^2$ step of Theorem 3.
- The decomposition of the total mass into low- and high-frequency parts, and the identity $S_k = X_k - X_{k-1}$ equal to the high-frequency mass, given the level-aggregation property as a hypothesis (Theorem 3).

- The exact rational values $S_1, \dots, S_6$ and their deviations from $7/15$ reported in Table 1, including the $k = 4, 5, 6$ rationals too large to print, each checked by exact arithmetic over $\mathbb{Q}$. The closed forms $S_1 = 2/3$ and $S_2 = 10/21$ are verified in the same file.

Cited, not re-derived in Lean. The level-aggregation property—that the level- k stationary distribution projects to the level- $(k - 1)$ distribution under reduction mod 3^{k-1} , so that the low-frequency mass of X_k equals X_{k-1} —is carried as a hypothesis in the formalization and is referenced to [2, Lemma 1.12]. The Plancherel decomposition (Theorem 3) is therefore verified *modulo* this cited input; the Fourier-analytic half of its proof is verified outright.

Not formalized. The identification of the surviving limit as exactly $S_\infty = 7/15$ is *certified* by the exact finite computation above, not proved as a Lean theorem; the verification confirms the computed values, not the passage to the limit. The rate of convergence (Section 7) is open and is not formalized.

What verification guarantees. A successful Lean check establishes that the stated propositions follow from `mathlib`’s axioms; it does not certify that the formal statements capture the paper’s intended objects. We have therefore stated each formalized result in the terms used above so that the correspondence is transparent. The verification’s role is narrow and specific: it removes any doubt that the Plancherel decomposition and the exact S_k data are correct as stated.

9 Connection to Tao’s Program

The constant $c = 7/45$ enters Tao’s measure-theoretic Collatz analysis [2] as the leading coefficient in the L^2 bound on Syracuse trajectory-measure increments, $|d_k|^2 \approx c \cdot (1/3)^k$. Determining its value sharpens that input. We emphasize what this paper does *not* supply: it does not establish the polynomial-in- A Fourier decay bound $|\hat{\mu}_n(\xi)| \lesssim_A n^{-A}$ that is the principal outstanding analytical step in Tao’s framework, nor the renewal/saddle machinery, nor the Collatz conjecture itself. It closes the value of one constant and isolates the rate question behind it.

10 Relation to the prefix-decomposition papers

This paper is part of a sequence on the residue-class structure of the Collatz map. The symbolic prefix decomposition and its per-class statistics, including the closed form $\langle \alpha_{\text{det}} \rangle = \log 6 / \log(4/3)$, are developed in Humphrey [19]; the structural origin of the empirical mod-8 stopping-time effect is treated in Humphrey [20]. The present paper concerns a different object - the high-frequency Plancherel mass of the Syracuse stationary measure - and depends on Humphrey [19] only for the prefix-statistics notation used in passing. The mechanism here (Plancherel $\times$ conservation $\times$ the diagonal operator) is self-contained.

A The exponential-sum obstruction for the convergence rate

Section 7 delimits the open rate question from the spectral side: no finite-rank operator over $\mathbb{Z}$ at finite truncation carries the sub-leading rate as a discrete eigenvalue. This appendix is the companion obstruction study referenced there, from the complementary character-sum side. The rigorous form of the rate is the square-root cancellation in the incomplete principal-unit sums of

[1, Prop. 20]; its validity is the short-window to that paper's unconditional shell-slice asymptotic [1, Thm. 26, Rem. 27]. We do not close it. We record which standard routes are obstructed, which are partially productive, and what remains, so that the rate is delimited from this side as sharply as Section 7 delimits it from the other.

A.1 The sum and the target

For $r \geq 1$, $\ell \in \{0, 1, 2\}$, $\varepsilon \in \{0, 1\}$, and $m \in \mathbb{Z}/N_r\mathbb{Z}$ with $N_r = 2 \cdot 3^{r-1}$,

$$S_{r,\ell,\varepsilon}(m) = \sum_{u=0}^{3^{r-1}-1} e_{3^{r+1}}(2^\varepsilon \omega_r^\ell 4^u - 9mu), \quad \widehat{h}_{r,\ell}(m) = \sum_{\varepsilon=0}^1 e_{N_r}(-m\varepsilon) S_{r,\ell,\varepsilon}(m). \quad (8)$$

The target [1, eq. (190)] is: for some $\delta > 0$ and all small $\eta > 0$,

$$\sup_{r \geq 1} \sup_{\ell} \sup_{\substack{J \subset \mathbb{Z}/N_r\mathbb{Z} \\ |J| \ll \sqrt{\eta} N_r}} N_r^{-1/2} \|\widehat{h}_{r,\ell}\|_{\ell^1(J)} \ll \eta^{1/2+\delta}. \quad (9)$$

The phase is non-stationary modulo 9 ([1, Prop. 21]: $v_3(\Phi') = 1$), so the obstruction is purely the incomplete-arc cancellation, not a stationary-phase contribution.

A.2 The character-sum route is partially productive

Applied with the constant-mod-3 phase polynomial (so the relevant degree parameter is $D = 0$), the Cochrane–Granville–Zheng character-sum machinery [17] is not a dead end: it yields two rigorous facts. Write $q = 3^{r+1}$ and $F(\xi) = \sum_{u=0}^{q-1} e_q(c 4^u - \xi u)$.

Proposition 7 (Complete-sum vanishing). *For every $r \geq 2$, ℓ , ε , m ,*

$$\sum_{u=0}^{3^{r+1}-1} e_{3^{r+1}}(c_{\ell,\varepsilon} 4^u - 9mu) = 0.$$

Proposition 8 (Fourier sparsity and mass). *$\widehat{F}$ is supported on $\{3a : a \in \mathbb{Z}/3^r\mathbb{Z}, a \equiv 1 \pmod{3}\}$, of cardinality $3^{r-1} = q/9$, and $|\widehat{F}(\xi)| = 3\sqrt{q}$ on that support.*

Both follow from the principal-unit structure of $1 + 3\mathbb{Z}_3$ together with Plancherel, and were verified by exact computation at $r = 2, 3, 4$. The support coincides with the $+$ class of the (P_+, P_-) decomposition of Section 5. Pólya–Vinogradov on this support gives only $|S_{\text{partial}}| \leq 3^r/3$, the trivial order N_r (and weaker than trivial for $r \geq 3$). The route therefore reduces (9) to a single residual estimate: a Burgess-strength mixed bilinear bound on $\sum_{a \equiv 1 \pmod{3}} \widehat{1}_N(3a) \widehat{F}(3a)$, where $\widehat{1}_N$ is the length- N_r window sum and $\widehat{F}$ carries the Gauss-sum magnitude of Proposition 8. The saving must come from phase cancellation in the *product* $\widehat{1}_N \cdot \widehat{F}$, not from either factor alone; that residual is precisely the class of incomplete mixed character sum treated by Cochrane et al. [17].

A.3 Van der Corput differencing is closed

Weyl differencing acts on $f(u) = e_q(c 4^u - 9mu)$ for a *single* m and yields a uniform-in- m magnitude bound. The $B=1$ Weyl- ℓ^1 form $|S|^2 \leq N + 2 \sum_{h>0} |A(h)|$ is a genuine sub-trivial bound; iterating to $B=2$ worsens it (Table 2).

Table 2: Empirical exponent rates for the differenced sums.

r	N	actual $ S $	$B=1$ rate	$B=2$ rate
3	9	4.74	0.769	0.798
4	27	10.33	0.750	0.807
5	81	17.46	0.715	0.809
6	243	32.34	0.727	0.809
7	729	55.76	0.725	—
8	2187	83.97	0.728	—

The $B=1$ rate stabilizes near 0.73 (rigorously sub-trivial; see the side product below); $B=2$ rises to ≈ 0.81 . Higher B approaches rate $1/2$ from above but never attains it (the process- B limit). The empirical rate of $|S|$ is $\approx 1/2$ but is not provable by differencing. Decisively, even the ideal pointwise bound does not suffice.

Proposition 9 (Pointwise insufficiency). *Suppose $|S_{r,c,m}| \leq C N_r^{1/2}$ for all m (the optimal pointwise rate). Then (9) still fails: every estimate of $\|\hat{h}_{r,\ell}\|_{\ell^1(J)}$ available from per- m information overshoots the target by a factor N_r^α with $\alpha > 0$.*

Proof sketch. With $|J| \leq C\sqrt{\eta} N_r$ and the granted bound: (i) pointwise, $\|\hat{h}\|_{\ell^1(J)} \leq |J| \sup |\hat{h}| \ll \sqrt{\eta} N_r^{3/2}$, so $N_r^{-1/2} \|\cdot\| \ll \sqrt{\eta} N_r$; (ii) Cauchy–Schwarz, $\leq \sqrt{|J|} \|\hat{h}\|_{\ell^2} \ll \eta^{1/4} N_r^{3/2}$, so $\eta^{1/4} N_r$; (iii) a uniform ℓ^2 -on-band hypothesis, $\ll \eta^{1/2+\delta} N_r^{3/2}$, so $\eta^{1/2+\delta} N_r$. Each carries a surplus factor N_r^α over the required $\eta^{1/2+\delta}$. Closing (9) therefore requires cancellation *between distinct* $m \in J$, which a single- m magnitude bound cannot supply. $\square$

A.4 A sub-trivial pointwise rate (side product)

The $B=1$ Weyl- ℓ^1 bound gives, rigorously and with no sum-product or smooth-modulus input,

$$|S_{r,c,m}| \leq C N_r^{0.73+o(1)}, \quad (10)$$

a genuine if modest improvement over the trivial $|S| \leq N_r$, through explicit control of the autocorrelations $A(h)$ via the lifting-the-exponent identity $v_3(4^h - 1) = 1 + v_3(h)$.

A.5 What remains

Both closed routes fail for the single reason isolated in Proposition 9: (9) is an off-diagonal cancellation statement in the m -variable, whereas Cochrane–Granville–Zheng and van der Corput supply only per- m magnitude control. The remaining live direction is the direct band- ℓ^1 analysis of $\hat{h}_{r,\ell}$ on the dangerous band $D_{r,t}(\eta)$ of [1, Prop. 22] — equivalently, (9) restated as a structural claim about the dual side — approachable via the joint law of $(j, \hat{h}_{r,\ell}(m))$ for m near t . This mirrors, on the character-sum side, the finite-rank spectral obstruction of Section 7: both show the sub-leading rate is real but unreachable by the standard per-mode or per- m tools, and both isolate the same residual object as the genuine open problem.

Table 3: Obstruction map for the rate- $\frac{1}{2}$ cancellation (9).

Route	Status	Outcome for (9)
Cochrane–Granville–Zheng	productive	Props. 7–8; residual Burgess bound
Pólya–Vinogradov	closed	weaker than trivial for $r \geq 3$
van der Corput $B=1$	closed	rate 0.73 (sub-trivial)
van der Corput $B=2$	closed	rate 0.81 (worse)
ideal pointwise $\sqrt{N}$	insufficient	Proposition 9
sum–product on $\langle 4 \rangle$	open	may reach $\sqrt{N}$; insufficient alone
smooth completion (aux. prime)	open	possible η^δ saving
direct band- ℓ^1 on $\hat{h}$	open	the actual requirement

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